

# Calculus Review I: Limits, Completeness, Asymptotic Scales, and Local Approximation

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## §1.1 Cauchy–Schwarz and Minkowski are error-control mechanisms

For vectors  $x, y \in \mathbb{R}^n$ , the Cauchy–Schwarz inequality is

$$|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2.$$

If  $y \neq 0$ , consider the nonnegative quadratic

$$0 \leq \|x - ty\|_2^2 = \|x\|_2^2 - 2t\langle x, y \rangle + t^2\|y\|_2^2.$$

Its discriminant is nonpositive, so

$$\langle x, y \rangle^2 \leq \|x\|_2^2 \|y\|_2^2.$$

Equality holds exactly when  $x$  and  $y$  are linearly dependent.

Minkowski’s inequality follows:

$$\begin{aligned} \|x + y\|_2^2 &= \|x\|_2^2 + 2\langle x, y \rangle + \|y\|_2^2 \\ &\leq \|x\|_2^2 + 2\|x\|_2 \|y\|_2 + \|y\|_2^2 \\ &= (\|x\|_2 + \|y\|_2)^2. \end{aligned}$$

Hence

$$\|x + y\|_2 \leq \|x\|_2 + \|y\|_2.$$

For nonzero vectors, equality holds exactly when  $x$  and  $y$  point in the same direction.

These inequalities control accumulated errors. If  $e_1, \dots, e_m$  are perturbations, then

$$\left\| \sum_{j=1}^m e_j \right\| \leq \sum_{j=1}^m \|e_j\|.$$

If an error is represented by an inner product, Cauchy–Schwarz converts it into a product of sizes. The same two patterns survive in Hilbert and normed spaces.

## §1.2 Limits are statements about eventual membership in neighborhoods

For  $a \in \mathbb{R}$ , let

$$N_\varepsilon(a) = \{x : |x - a| < \varepsilon\}.$$

A sequence  $(x_n)$  converges to  $a$  when

$$\forall \varepsilon > 0, \quad x_n \in N_\varepsilon(a) \text{ eventually.}$$

Here “eventually” means that there exists  $N$  such that the statement holds for every  $n \geq N$ .

For a function  $f$ ,

$$\lim_{x \rightarrow x_0} f(x) = L$$

means that for every neighborhood  $V$  of  $L$ , there is a punctured neighborhood  $U$  of  $x_0$  such that

$$x \in U \implies f(x) \in V.$$

In epsilon–delta notation,

$$0 < |x - x_0| < \delta \implies |f(x) - L| < \varepsilon.$$

The neighborhood families form filters: intersections of finitely many neighborhoods are neighborhoods, and larger sets remain neighborhoods. This is why limits compose. If

$$f(x) \rightarrow L \quad \text{and} \quad g(y) \rightarrow M \quad \text{as} \quad y \rightarrow L,$$

then the preimage under  $g$  of a neighborhood of  $M$  contains a neighborhood of  $L$ , and the preimage under  $f$  of that neighborhood contains a punctured neighborhood of  $x_0$ .

## §1.3 Sequence limits are unique and respect algebraic operations

Suppose  $x_n \rightarrow a$  and  $x_n \rightarrow b$ . If  $a \neq b$ , choose

$$\varepsilon = \frac{|a - b|}{3}.$$

For large  $n$ ,

$$|a - b| \leq |a - x_n| + |x_n - b| < 2\varepsilon < |a - b|,$$

a contradiction. Thus limits are unique.

If  $x_n \rightarrow a$  and  $y_n \rightarrow b$ , then

$$x_n + y_n \rightarrow a + b, \quad x_n y_n \rightarrow ab.$$

For the product,

$$|x_n y_n - ab| \leq |x_n| |y_n - b| + |b| |x_n - a|.$$

A convergent sequence is bounded, so  $|x_n| \leq M$  eventually, and both terms tend to zero. If  $b \neq 0$ , then  $y_n \neq 0$  eventually and

$$\frac{x_n}{y_n} \rightarrow \frac{a}{b}.$$

The definition of  $x_n \rightarrow +\infty$  is also eventual:

$$\forall M > 0, \quad x_n > M \text{ eventually.}$$

It is not convergence to a real number, but it becomes ordinary convergence after adjoining points at infinity or applying a compactifying coordinate such as  $x \mapsto \arctan x$ .

## §1.4 Heine's criterion converts function limits into sequence limits

### Theorem 1.4.1 (Heine criterion)

Let  $f$  be defined on a punctured neighborhood of  $x_0$ . Then

$$\lim_{x \rightarrow x_0} f(x) = L$$

iff for every sequence  $x_n \rightarrow x_0$  with  $x_n \neq x_0$ , one has

$$f(x_n) \rightarrow L.$$

*Proof.* Assume the epsilon-delta limit. Given  $\varepsilon > 0$ , choose  $\delta > 0$  so that

$$0 < |x - x_0| < \delta \implies |f(x) - L| < \varepsilon.$$

Since  $x_n \rightarrow x_0$ , the hypothesis on  $x_n$  holds eventually, so  $f(x_n) \rightarrow L$ .

Conversely, suppose the epsilon-delta condition fails. Then there exists  $\varepsilon_0 > 0$  such that for every  $n$  one can choose  $x_n$  with

$$0 < |x_n - x_0| < \frac{1}{n}, \quad |f(x_n) - L| \geq \varepsilon_0.$$

Then  $x_n \rightarrow x_0$ , but  $f(x_n) \not\rightarrow L$ , contradicting the sequential assumption.  $\square$

To disprove a limit, construct incompatible approaches. For

$$f(x) = \cos \frac{1}{x},$$

take

$$x_n = \frac{1}{2\pi n}, \quad y_n = \frac{1}{(2n+1)\pi}.$$

Both tend to zero, but

$$f(x_n) = 1, \quad f(y_n) = -1.$$

Therefore  $\lim_{x \rightarrow 0} f(x)$  does not exist.

## §1.5 Asymptotic notation compares scales, not fixed small numbers

Let  $g(x) > 0$  near a limiting point. The notation

$$f = O(g)$$

means that there are constants  $C > 0$  and a neighborhood on which

$$|f(x)| \leq Cg(x).$$

The notation

$$f = o(g)$$

means

$$\frac{f(x)}{g(x)} \rightarrow 0.$$

We write

$$f = \Theta(g)$$

when both  $f = O(g)$  and  $g = O(f)$ , and

$$f \sim g$$

when

$$\frac{f(x)}{g(x)} \rightarrow 1.$$

The implications are

$$f \sim g \implies f = \Theta(g), \quad f = o(g) \implies f = O(g),$$

but the converses fail.

Equivalent infinitesimals may be substituted safely in products and quotients when the remaining factors have controlled limits. For example,

$$\sin x \sim x, \quad 1 - \cos x \sim \frac{x^2}{2} \quad (x \rightarrow 0).$$

Hence

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x} = \lim_{x \rightarrow 0} \frac{x^2/2}{x^2} = \frac{1}{2}.$$

Substitution is not valid inside arbitrary sums where cancellation may change the leading order. Although

$$\sqrt{1+x} \sim 1,$$

one cannot replace it by 1 in

$$\sqrt{1+x} - 1.$$

The difference is of smaller scale:

$$\sqrt{1+x} - 1 = \frac{x}{\sqrt{1+x} + 1} \sim \frac{x}{2}.$$

## §1.6 Squeeze and monotone convergence use order structure

The squeeze theorem says that if

$$a_n \leq b_n \leq c_n$$

eventually and

$$a_n \rightarrow L, \quad c_n \rightarrow L,$$

then  $b_n \rightarrow L$ . Indeed,

$$|b_n - L| \leq \max\{|a_n - L|, |c_n - L|\}$$

when  $L$  lies between the bounding terms.

A monotone bounded sequence converges because  $\mathbb{R}$  has suprema.

### Theorem 1.6.1 (Monotone convergence)

If  $(x_n)$  is increasing and bounded above, then

$$x_n \rightarrow \sup_n x_n.$$

*Proof.* Let

$$L = \sup\{x_n : n \in \mathbb{N}\}.$$

Given  $\varepsilon > 0$ , the number  $L - \varepsilon$  is not an upper bound. Hence some  $N$  satisfies

$$x_N > L - \varepsilon.$$

For  $n \geq N$ , monotonicity gives

$$L - \varepsilon < x_N \leq x_n \leq L.$$

Thus  $|x_n - L| < \varepsilon$ . □

The proof exposes the completeness input: without the least-upper-bound property, the candidate  $L$  may be missing from the number system.

## §1.7 The Cauchy criterion characterizes convergence in complete spaces

A sequence is Cauchy if

$$\forall \varepsilon > 0, \quad |x_m - x_n| < \varepsilon \text{ for all sufficiently large } m, n.$$

Every convergent sequence is Cauchy by the triangle inequality.

For the converse in  $\mathbb{R}$ , first note that a Cauchy sequence is bounded. Choose  $N$  so that

$$|x_n - x_N| < 1 \quad (n \geq N).$$

The tail lies in a bounded interval, and the finitely many earlier terms are also bounded.

By Bolzano–Weierstrass, a bounded sequence has a convergent subsequence

$$x_{n_k} \rightarrow L.$$

Given  $\varepsilon > 0$ , choose  $K$  so that

$$|x_m - x_n| < \frac{\varepsilon}{2} \quad (m, n \geq K)$$

and choose  $k$  with  $n_k \geq K$  and

$$|x_{n_k} - L| < \frac{\varepsilon}{2}.$$

Then for every  $n \geq K$ ,

$$|x_n - L| \leq |x_n - x_{n_k}| + |x_{n_k} - L| < \varepsilon.$$

Thus  $x_n \rightarrow L$ .

The same statement fails in  $\mathbb{Q}$ . Let

$$q_n = \frac{\lfloor 10^n \sqrt{2} \rfloor}{10^n}.$$

Then  $(q_n)$  is Cauchy in the rational metric, but it has no limit in  $\mathbb{Q}$ . Completion adds the missing limit  $\sqrt{2}$ .

A Banach space is exactly a normed vector space in which every Cauchy sequence converges. Completeness is therefore not a technical appendix; it is what turns internal error control into the existence of a limit.

## §1.8 Continuity at each point is weaker than uniform continuity

A function  $f : X \rightarrow Y$  between metric spaces is uniformly continuous if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon$$

for all  $x, y \in X$ . The same  $\delta$  must work everywhere.

The function  $f(x) = x^2$  is continuous on  $\mathbb{R}$  but not uniformly continuous. Set

$$x_n = n, \quad y_n = n + \frac{1}{n}.$$

Then

$$|x_n - y_n| = \frac{1}{n} \rightarrow 0,$$

but

$$|x_n^2 - y_n^2| = 2 + \frac{1}{n^2} \not\rightarrow 0.$$

On compact domains, continuity becomes uniform.

### Theorem 1.8.1 (Heine–Cantor)

A continuous function from a compact metric space to a metric space is uniformly continuous.

*Proof.* If uniform continuity failed, there would be  $\varepsilon_0 > 0$  and pairs  $x_n, y_n$  with

$$d(x_n, y_n) < \frac{1}{n}, \quad d(f(x_n), f(y_n)) \geq \varepsilon_0.$$

Compactness gives a convergent subsequence  $x_{n_k} \rightarrow x$ . Since

$$d(y_{n_k}, x) \leq d(y_{n_k}, x_{n_k}) + d(x_{n_k}, x) \rightarrow 0,$$

one also has  $y_{n_k} \rightarrow x$ . Continuity then forces both image subsequences to converge to  $f(x)$ , contradicting their separation by  $\varepsilon_0$ .  $\square$

Uniform continuity preserves Cauchy sequences and is therefore the natural hypothesis for extending a map to a completion.

## §1.9 A derivative is a first-order approximation with a controlled remainder

The derivative at  $a$  can be written as

$$f(a + h) = f(a) + f'(a)h + r(h), \quad \frac{r(h)}{h} \rightarrow 0.$$

Equivalently,

$$f(a + h) = f(a) + f'(a)h + o(h).$$

This is stronger than the existence of a slope formula: it says the nonlinear error is negligible compared with the increment.

The differential

$$df_a(h) = f'(a)h$$

is the linear map giving the best first-order approximation. In several variables, the correct extension is the Fréchet derivative:

$$f(a + h) = f(a) + Df(a)h + o(\|h\|).$$

The derivative is therefore a local linear model, not merely a quotient of small numbers.

The chain rule follows by composing the two local models. If

$$g(a + h) = g(a) + g'(a)h + o(h)$$

and

$$f(g(a) + k) = f(g(a)) + f'(g(a))k + o(k),$$

then  $k = g'(a)h + o(h) = O(h)$ , so

$$f(g(a + h)) = f(g(a)) + f'(g(a))g'(a)h + o(h).$$

Thus

$$(f \circ g)'(a) = f'(g(a))g'(a).$$

## §1.10 Taylor's theorem is a finite jet plus a quantified remainder

If  $f \in C^{n+1}$  near  $a$ , Taylor's theorem gives

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k + R_n(x).$$

The Lagrange remainder has the form

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - a)^{n+1}$$

for some  $\xi$  between  $a$  and  $x$ .

The integral remainder is

$$R_n(x) = \frac{1}{n!} \int_a^x f^{(n+1)}(t) (x - t)^n dt.$$

It follows by repeated use of the fundamental theorem of calculus. For example,

$$\begin{aligned} f(x) - f(a) &= \int_a^x f'(t_1) dt_1 \\ &= f'(a)(x - a) + \int_a^x \int_a^{t_1} f''(t_2) dt_2 dt_1, \end{aligned}$$

and exchanging the order of integration produces the kernel  $(x - t)$ . Iterating gives the general formula.

If

$$|f^{(n+1)}(t)| \leq M$$

between  $a$  and  $x$ , then

$$|R_n(x)| \leq \frac{M|x - a|^{n+1}}{(n+1)!}.$$

For  $e^x$  on  $[0, 1]$ , the degree- $n$  Maclaurin approximation satisfies

$$\left| e^x - \sum_{k=0}^n \frac{x^k}{k!} \right| \leq \frac{e}{(n+1)!}.$$

Taylor expansion is therefore an asymptotic statement with a certified error, not permission to replace every smooth function by an infinite series.

## §1.11 Rotation matrices generate trigonometric identities

Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = -I.$$

The matrix exponential separates into even and odd powers:

$$\begin{aligned} e^{tJ} &= I \cos t + J \sin t \\ &= \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}. \end{aligned}$$

It preserves the Euclidean quadratic form:

$$(e^{tJ})^T e^{tJ} = I.$$

The group law

$$e^{(s+t)J} = e^{sJ} e^{tJ}$$

gives the addition formulas by matrix multiplication:

$$\cos(s+t) = \cos s \cos t - \sin s \sin t,$$

$$\sin(s+t) = \sin s \cos t + \cos s \sin t.$$

Thus the identities are coordinates of the rotation group, not an isolated table to memorize.

## §1.12 Lorentz boosts generate hyperbolic identities

Let

$$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad K^2 = I.$$

Then

$$\begin{aligned} e^{tK} &= I \cosh t + K \sinh t \\ &= \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix}. \end{aligned}$$

Let

$$\eta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

A direct multiplication gives

$$(e^{tK})^T \eta e^{tK} = \eta.$$

Therefore the boost preserves

$$x^2 - y^2.$$

Applying it to  $(1, 0)^T$  yields

$$\begin{pmatrix} \cosh t \\ \sinh t \end{pmatrix},$$

so invariance gives

$$\boxed{\cosh^2 t - \sinh^2 t = 1.}$$



Again the group law

$$e^{(s+t)K} = e^{sK}e^{tK}$$

produces

$$\cosh(s+t) = \cosh s \cosh t + \sinh s \sinh t,$$

$$\sinh(s+t) = \sinh s \cosh t + \cosh s \sinh t.$$

Circular and hyperbolic functions are parallel because both arise from one-parameter matrix groups. The sign difference comes from

$$J^2 = -I, \quad K^2 = I,$$

and from the quadratic forms they preserve: Euclidean  $x^2 + y^2$  versus Lorentzian  $x^2 - y^2$ .